Informatica

Informatica® Cloud Data Integration

SAP BW BEx Query Connector

Informatica Cloud Data Integration SAP BW BEx Query Connector
April 2026

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Table of Contents

Preface 4

Informatica Resources 4

- Informatica Documentation 4
- IDMC web site 4
- IDMC Communities 4
- IDMC Marketplace 4
- Data Integration connector documentation 5
- Informatica Knowledge Base 5
- IDMC Trust Center 5
- Informatica Global Customer Support 5

Chapter 1: Introduction to SAP BW BEx Query Connector 6

SAP BW BEx Query Connector assets 6

Chapter 2: Connections for SAP BW BEx Query 7

Prerequisites 7

- Download and configure the SAP libraries 7
- Configure SAP user authorization 8

Connect to SAP BW BEx Query 9

- Before you begin 9
- Connection details 9
- Connection types 10

Use the serverless runtime environment 13

Chapter 3: Mappings and mapping tasks with SAP BW BEx 15

Prerequisites 15

SAP BW BEx sources in mappings 16

Advanced source properties in mappings 16

Configuring a mapping with a single SAP BW BEx source 16

Creating a mapping task 18

Monitoring a mapping task 19

Mapping and mapping task example with SAP BW BEx Query 19

Rules and guidelines for SAP BW BEx Query operations 23

Limitations for SAP BW BEx Query operations 23

Appendix A: SAP data type reference 24

SAP BW BEx Transformation Data Types 24

Index 26

Preface

Use *SAP BW BEx Query Connector* to learn how to read from SAP BEx queries by using Cloud Data Integration. Learn to create an SAP BW BEx Query connection, develop and run mappings and tasks in Cloud Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

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https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

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CHAPTER 1

Introduction to SAP BW BEx Query Connector

You can use SAP BW BEx Query Connector to read data from SAP BEx queries. SAP Connector uses the SAP APIs to integrate with the SAP BW system and read data from BEx queries.

You can use the SAP BW BEx Query connection in mappings and mapping tasks. Create a mapping task to process data based on the data flow logic defined in a mapping or integration template.

You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality.

Note: Informatica recommends that you do not use SAP BW BEx Query Connector to extract large volumes of data. SAP BW BEx Query Connector was primarily designed to extract complex calculated fields from SAP where the logic cannot be replicated outside. You must consider extracting only the key fields and the calculated fields through the BEx query and merge with the rest of the dimensional data outside in a data warehouse.

SAP BW BEx Query Connector assets

Create assets in Data Integration to integrate data using SAP BW BEx Query Connector.

When you use SAP BW BEx Query Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for SAP BW BEx Query

Create an SAP BW BEx Query connection to securely read SAP BW BEx queries from SAP BW.

You can use an SAP BW BEx Query connection in mappings and mapping tasks.

Prerequisites

Before you use an SAP BW BEx Query connection, the SAP administrator needs to perform certain prerequisite tasks to configure the Secure Agent machine and SAP system.

To process SAP BW BEx Query data, you also need to verify if the required licenses are enabled for the SAP system.

Download and configure the SAP libraries

To read data from SAP BW BEx Query, you need to download and configure the SAP JCo libraries on the Secure Agent machine. If you encounter any issues while you download libraries, contact SAP Customer Support.

1. Go to the SAP Support Portal, and then click **Software Downloads**.

   Note: You need to have SAP credentials to access **Software Downloads** from the SAP Support Portal.
2. Download the latest version of the 64-bit SAP JCo libraries based on the operating system of the machine on which the Secure Agent runs.

Operating System	SAP JCo Libraries
Windows	- sapjco3.jar - sapjco3.dll
Linux	- sapjco3.jar - libsapjco3.so

3. Copy the JCo libraries to the following directory:
   `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\ext\deploy_to_main\bin\rdtm-extra\tpl\sap`

   Create the `deploy_to_main\bin\rdtm-extra\tpl\sap` directory if it does not already exist.

4. Log in to Informatica® Intelligent Data Management Cloud™ (IDMC) and configure the JAVA_LIBS property for the Secure Agent.
   a. Select **Administrator** > **Runtime Environments**.
   b. Click **Runtime Environments** to access the **Runtime Environments** page.
   c. To the left of the agent name, click **Edit Secure Agent**.
   d. From the **Service** list, select **Data Integration Server**.
   e. From the **Type** list, select **Tomcat JRE**.
   f. Enter the JAVA_LIBS value based on the operating system of the machine on which the Secure Agent runs.

Operating System	Value
Windows	..\bin\rdtm-extra\tpl\sap\sapjco3.jar;..\bin\rdtm\javalib\sap\sap-adapter-common.jar
Linux	../bin/rdtm-extra/tpl/sap/sapjco3.jar:../bin/rdtm/javalib/sap/sap-adapter-common.jar

   Warning: If you copy the value directly from the table, the hyphens (-) in the value might be incorrectly copied. Copy the value to a text editor and make sure that the value you copied is not corrupted.



   g. Click **Save**.
   h. Repeat steps 2 through 4 on every machine where you installed the Secure Agent.
5. Restart the Secure Agent.

Configure SAP user authorization

Configure the SAP user account in the SAP system to process SAP BW BEx Query data.

The following table lists the objects and authorization required to configure user permissions to access and read data from SAP BW BEx Query:

Read Object Name	Authorization
S_RFC	RFC1, RFC_METADATA, RFC_METADATA_GET, RSAB, RSOB, RS_UNIFICATION, SDTX, SUGU, SU_USER, SYST
S_RS_COMP	ACTVT=3 (DISPLAY)
S_RS_COMP1	ACTVT=3 (DISPLAY)
S_RS_ICUBE	ACTVT=3 (DISPLAY)

For more information about these objects and how to use them, see SAP user authorizations.

Connect to SAP BW BEx Query

Let's configure the SAP BW BEx Query connection properties to connect to SAP BW and read data from SAP BW BEx Query.

Before you begin

Before you get started, you'll need to configure the Secure Agent machine and SAP system to establish an SAP BW BEx Query connection.

Check out "Prerequisites" on page 7 to learn more about these tasks.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, serverless, or elastic runtime environment. For more information about how to configure a serverless environment, see "Use the serverless runtime environment" on page 13. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
Authentication	The authentication type to access SAP BW and read SAP BW BEx queries. Select the **SAP** authentication type and then configure the authentication-specific parameters.
Username	The user name with the appropriate user authorization to connect to the SAP BW account.
Password	The password to connect to the SAP BW account.

Connection types

You can configure application and load balancing connection types to connect to SAP BW and read data from SAP BW BEx Query.
Select the required connection type and then configure the connection-specific parameters.

Application connection

Application connection is the default type which requires your SAP account name, password, client number, host name, system number, and language code.

The following table describes the basic connection properties for an application connection:

Connection property	Description
Host name	The host name or IP address of the SAP BW server to which you want to connect.
System number	The system number of the SAP BW server. Get the required system number from the SAP system to which you want to connect.
Client	The client number of the SAP BW server. Get the required client number from the SAP system to which you want to connect.
Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.

Advanced settings

The following table describes the advanced connection properties for an application connection:

Property	Description
Trace	Determines whether to track the JCo calls that the SAP system makes. Enter one of the values between 0 and 8. By default, SAP doesn't store information about the JCo calls in a trace file. Default is 0. For more information about trace level values, see Setting up an SAP Java Connector (SAP JCo) and Related Traces. You can access the trace files from the following directories: - Design time information: `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<Latest version>\ICS\main\tomcat` - Run-time information: `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<Latest version>\ICS\main\bin\rdtm`

Additional parameters	Additional JCo connection parameters that you can use when you read SAP BW BEx queries from SAP BW.
	You can enter multiple JCo connection parameters, separated by a semicolon, in the following format:
	`<parameter name1>=<value1>;<parameter name2>=<value2>;<parameter name3>=<value3>....`
	For example, if you want to create an SAP SNC connection, enter the following additional parameters:
	`SNC_MODE=1;SNC_QOP=3;SNC_MYNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>, C=<Country>;SNC_PARTNERNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>, C=<Country>;SNC_LIB=C:\SNC_11\sapcrypto.dll;TRACE=1`

SAP Additional Parameters	Additional SAP properties that the Secure Agent uses to connect to the SAP system as an RFC client.
	Specify the required RFC-specific parameters and connection information that enable the Secure Agent to connect to SAP.
	For example, you can specify the SNC connection parameters as additional arguments listed in the following sample:
	`GROUP=interfaces`
	`ASHOST=<Application server hostname>`
	`SYSNR=20`
	`SNC_MODE=1`
	`SNC_PARTNERNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>`
	`SNC_MYNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>`
	`SNC_LIB=<Secure Agent installation directory>/apps/server/bin/<libsapcrypto.so for Linux/sapcrypto.dll for Windows>`
	`X509CERT=<X509 certificate>`
	`TRACE=2`
	For more information about RFC-specific parameters, see the SAP documentation.
	Note: If you specify a parameter in both the **SAP Additional Parameters** and **Additional parameters** fields, the value specified in the **SAP Additional Parameters** field takes precedence.

Load balancing connection

Create a load balancing connection when you want to connect to the SAP BW server with the least load at run time.

When you create a load balancing connection, in addition to all the properties listed in the **Connection Details** section, you also need to enter the message host name, R3 name/SysID, and group values in the **Advanced Settings** section.

The following table describes the basic connection properties for a load balancing connection:

Connection property	Description
Username	The user name with the appropriate user authorization to connect to the SAP BW account.
Password	The password to connect to the SAP BW account.
Host name	The host name or IP address of the SAP BW server to which you want to connect.
System number	The system number of the SAP BW server. Get the required system number from the SAP system to which you want to connect.
Client	The client number of the SAP BW server. Get the required client number from the SAP system to which you want to connect.
Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.

Advanced settings

The following table describes the advanced connection properties for a load balancing connection:

Property	Description
Message host name	Required. The host name of the SAP message server to which you want to connect when you use a load balancing connection.
R3 name/ SysID	Required. The system ID of the SAP message server to which you want to connect when you use a load balancing connection.
Group	Required. The name of the SAP logon group through which you want to connect when you use a load balancing connection.
Trace	Determines whether to track the JCo calls that the SAP system makes. Enter one of the values between 0 and 8. By default, SAP doesn't store information about the JCo calls in a trace file. Default is 0. For more information about trace level values, see Setting up an SAP Java Connector (SAP JCo) and Related Traces. You can access the trace files from the following directories: - Design time information: `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<Latest version>\ICS\main\tomcat` - Run-time information: `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<Latest version>\ICS\main\bin\rdtm`

Property	Description

Additional parameters	Additional JCo connection parameters that you can use when you read SAP BW BEx queries from SAP BW.
	You can enter multiple JCo connection parameters, separated by a semicolon, in the following format:
	`<parameter name1>=<value1>;<parameter name2>=<value2>;<parameter name3>=<value3>....`
	For example, if you want to create an SAP SNC connection, enter the following additional parameters:
	`SNC_MODE=1;SNC_QOP=3;SNC_MYNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>, C=<Country>;SNC_PARTNERNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>, C=<Country>;SNC_LIB=C:\SNC_11\sapcrypto.dll;TRACE=1`

SAP Additional Parameters	Additional SAP properties that the Secure Agent uses to connect to the SAP system as an RFC client.
	Specify the required RFC-specific parameters and connection information that enable the Secure Agent to connect to SAP.
	For example, you can specify the SNC connection parameters as additional arguments listed in the following sample:
	`GROUP=interfaces`
	`ASHOST=<Application server hostname>`
	`SYSNR=20`
	`SNC_MODE=1`
	`SNC_PARTNERNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>`
	`SNC_MYNAME=p:CN=<Common name>, OU=<Organizational unit>, O=<Organization>`
	`SNC_LIB=<Secure Agent installation directory>/apps/server/bin/<libsapcrypto.so for Linux/sapcrypto.dll for Windows>`
	`X509CERT=<X509 certificate>`
	`TRACE=2`
	For more information about RFC-specific parameters, see the SAP documentation.
	Note: If you specify a parameter in both the **SAP Additional Parameters** and **Additional parameters** fields, the value specified in the **SAP Additional Parameters** field takes precedence.

Use the serverless runtime environment

You can use a serverless runtime environment hosted on Azure to connect to the SAP system when you configure an SAP BW BEx Query connection on Linux.

You can't use the serverless runtime environment if you want to use SAP Secure Network Communication (SNC) Protocol.

Before you configure an SAP BW BEx Query connection using the serverless runtime environment, perform the following tasks:

- Add the SAP libraries in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.
- Configure the JAVA_LIBS property for the serverless runtime environment on Linux.

Add the SAP libraries in the Azure container in your Azure account

Perform the following steps to configure an SAP BW BEx Query connection in a serverless runtime environment:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config`
2. Add the SAP libraries in the Azure container in the following location in your Azure account: `<Supplementary file location>/serverless_agent_config/sap`

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment, and to copy the SAP libraries to the serverless agent directory:

1. Copy the following code snippet to a text editor:

```
version: 1
agent:
  dataIntegrationServer:
    autoDeploy:
      sap:
        nwrfcs:
          - fileCopy:
              sourcePath: sap/nwrfc/<rfc_libary_filename>
          - fileCopy:
              sourcePath: sap/nwrfc/<sapnwrfc_filename>
```

   where the source path is the directory path of the SAP library files in Azure.
2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the SAP libraries are copied from the Azure location to the serverless agent directory.

Configure the JAVA_LIBS property for the serverless runtime environment

Perform the following steps in Administrator to configure the JAVA_LIBS property for the serverless runtime environment on Linux:

1. Log in to Informatica® Intelligent Data Management Cloud™ (IDMC).
2. Select **Administrator** > **Serverless Environments**.
3. On the **Serverless Environments** tab, expand the Actions menu for the required serverless runtime environment, and then select **Edit**.
4. On the **Runtime Configuration Properties** tab, select **Data Integration Server** as the service and **Tomcat_JRE** as the type.
5. Click **Add Property**.
6. Enter JAVA_LIBS in the **Name** field and set the following value: `../bin/rdtm-extra/tpl/sap/sapjco3.jar:../bin/rdtm/javalib/sap/sap-adapter-common.jar`
7. Click **Save**.

For more information about how to configure and use the serverless environment, see "Serverless runtime environment setup" in *Runtime Environments* in the Administrator help.

CHAPTER 3

Mappings and mapping tasks with SAP BW BEx

You can create a mapping to read data from SAP BEx queries. SAP Connector uses the SAP APIs to integrate with the SAP BW system and read data from BEx queries.

You can override the source connection at run time with values specified in a parameter file.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

When you configure a mapping to describe the flow of data from a source to a target, you can also add transformations to transform data. A transformation includes field rules to define incoming fields. Links visually represent how data moves through the data flow.

After you create a mapping, you can run the mapping or you can deploy the mapping in a mapping task. You can use the Mapping application to process data based on the data flow logic that you defined in the mapping.

Use the Mapping Task wizard to create a mapping task. When you create a mapping task, you can select the mapping based on which you want to create the task. You can associate the task with a schedule to run it at specified times or at regular intervals. Or, you can run it manually. You can monitor tasks and view details about completed tasks.

Prerequisites

Before you use SAP Connector to read data from an SAP BEx query, you must allow access to the SAP BEx query from external systems.

1. Open the query in the SAP BEx Query Designer.
2. Click the **Extended** tab on the right pane.
3. In the **Release for External Access** section, select the **By OLE DB for OLAP** option under **Allow External Access to this Query**.

SAP BW BEx sources in mappings

To read data from an SAP BEx query, configure a Source transformation in the mapping.

When you select an SAP BW BEx query connection for a Source transformation, you can select a read operation that represents the imported SAP BEx query from which you want to read data.

You can use the sample XML template provided to configure the request message in an XML format. You can also configure filter conditions in the request XML to filter the data that you read from an SAP BEx query. Configure the advanced properties to set the tracing level and the cache size for the Web service response.

You can view the response structure in the field mapping. When you map the elements from the response structure to the output fields, the Secure Agent creates an output group. Map the fields from the SAP BEx query to the output group, and then connect the output group to a transformation or target to which you want to write data.

Advanced source properties in mappings

In a mapping, you can configure a Source transformation to represent an SAP BEx query source. For SAP BW BEx query connections used in mapping tasks, you can configure advanced properties in the **Source** page of the Mapping Task wizard.

The following table describes the advanced properties that you can configure in a Source transformation:

Property	Description
Tracing Level	Amount of detail that appears in the log for the Source transformation. Use the following tracing levels: - Terse - Normal - Verbose Initialization - Verbose Default is normal.
Cache Size for Web Service Response (KB)	Memory available for the web service response. The cache size depends on the amount of data you read from the SAP BEx query. If the web service response contains many rows or columns, you might want to increase the cache size. You can specify a maximum size of 99999 KB. Default is 1024 KB.

Configuring a mapping with a single SAP BW BEx source

Use the Data Integration Mapping Designer to configure a mapping. Create a Source transformation to read data from an SAP BW BEx query.

1. To create a mapping, click **Data Integration** > **New** > **Mappings**. Select **Mapping** and click **Create**.

2. Enter a name and description for the mapping, and click **OK**.

   You can use alphanumeric characters and underscores (_) in the mapping name.
3. To configure an SAP BW BEx query source, on the **Transformation** palette, click **Source**.
4. In the **Properties** panel, on the **General** tab, enter a name and description for the SAP BW BEx query source.
5. Click the **Source** tab and configure the SAP BW BEx query source details.
   a. In the **Connection** list, specify the SAP BW BEx query connection that you want to use to read data.
   b. In the **Operation** list, click **Select** to specify the read operation that you want to use to read data. When you click **Select**, a list of SAP BEx queries appears. You can search for a BEx query based on the technical name of the query. Special characters in the query are replaced with _. Select the SAP BEx query from which you want to read data.
   c. Under the **Request Options** section, click **Configure**. The **Edit Request Message** page appears and displays a sample XML template that you can use for the request message. You can copy and paste the sample message into the **Request Message** text box or customize the sample message. For example, if you want to filter the data that you read from the SAP BEx query, you can add filter conditions in the request message. Click **OK** to save the request message.
   d. Under the **Advanced** section, configure the tracing level and the cache size for the Web service response.
6. In the **Field Mapping** tab, click **Map all descendants** to map all the incoming fields from the SAP BEx query to the fields in the output group.
7. Click the **Fields** tab to edit the precision and scale for the output fields.

   Note: The Secure Agent imports the decimal data type as double. You must manually edit the scale and precision of decimal fields so that they match with the values in SAP. The maximum precision that SAP BW BEx Query Connector supports for key figure data is 22. If the precision exceeds 22, the data gets corrupted.
8. To add a transformation, on the **Transformation palette**, click the transformation name. Or, drag the transformation onto the mapping canvas.
   a. On the **General** tab, you can enter a name and description for the transformation.
   b. Draw a link to connect the output group to the transformation.

      When you link transformations, the downstream transformation inherits the incoming fields from the previous transformation.
      For a Joiner transformation, draw a master link and a detail link.
   c. To preview fields, configure the field rules, or rename fields, click **Incoming Fields**.
   d. Configure additional transformation properties, as needed.

      The properties that you configure vary based on the type of transformation you create.
   e. To add another transformation, repeat these steps.
9. To add a Target transformation, on the **Transformation palette**, click **Target**.
   a. On the **General** tab, enter a name and description for the target.
   b. Draw a link to connect the previous transformation to the Target transformation.
   c. Click the **Target** tab and configure target details. You can also create a target at run time. If necessary, configure the advanced target properties.

      Target details and advanced target properties appear based on the connection type.
   d. To preview fields, configure the field rules, or rename fields, click **Incoming Fields**.

e. Click **Field Mapping** and map the fields that you want to write to the target.

f. To add another Target transformation, repeat these steps.

10. Save the mapping. You can choose one of the following save options:
    - Click **Save and Close** to save and close the mapping.
    - Click **Save and Run** to save and run the mapping immediately.
    - Click **Save and New Mapping Task** to save and create a mapping task.

Creating a mapping task

You can create a mapping task based on a valid mapping.

1. To create a mapping task, click **Data Integration** > **New** > **Tasks** and then complete one of the following steps:
    - To create a mapping task based on a mapping, select **Mapping Task** and click **Create**.
    - To create a mapping task using a template, expand the appropriate template category and select the template you want to use, and then click **Create**.

    To edit a mapping task, on the **Explore** page, navigate to the mapping task. In the row that contains the task, click **Actions** and select **Edit**.
2. Enter a name for the task.

    Task names must be unique within the organization. Task names are not case sensitive. Task names can contain alphanumeric characters, spaces, and the following special characters: _ . + -
3. Select the runtime environment that contains the Secure Agent that you want to use to access the SAP BEx queries.
4. Select **Mapping** as the task based on which you want to create the mapping task.
5. Click **Select** to specify a mapping.

    The **Select a Mapping** dialog box appears.
6. Select a mapping. You can also search for a mapping based on its name and description.
7. Click **OK**.

    The image of the selected mapping appears.
8. Click **Next**.

    The **Runtime Options** page appears.
9. Configure a schedule and advanced options. Perform any of the following steps based on your requirements:
    a. Click **Run this task on schedule** and specify the schedule that you want to use.

    b. Configure the email notification options.

    c. Configure advanced options for the task.

    d. Specify the execution mode.
10. Optionally, add advanced session properties.
    a. Click **Add**.

    b. Select a session property.

c. Configure the value of the session property.

11. Save the mapping task. You can choose one of the following save options:
    - Click **Save and Close** to save and close the task.
    - Click **Save and Continue** to save the task and continue with configuring the task.
    - Click **Save and Run** to save and run the task immediately.

Monitoring a mapping task

After you run a mapping task, you can monitor the task and view the logs.

In **Monitor**, you can monitor the status of the logs after you run the task.

You can also monitor the progress of the task by calling Transaction SM37 from SAP. You can view the actual job duration in SAP. The job duration listed in the Data Integration activity log is a higher value because it also includes time required to complete processing in Data Integration.

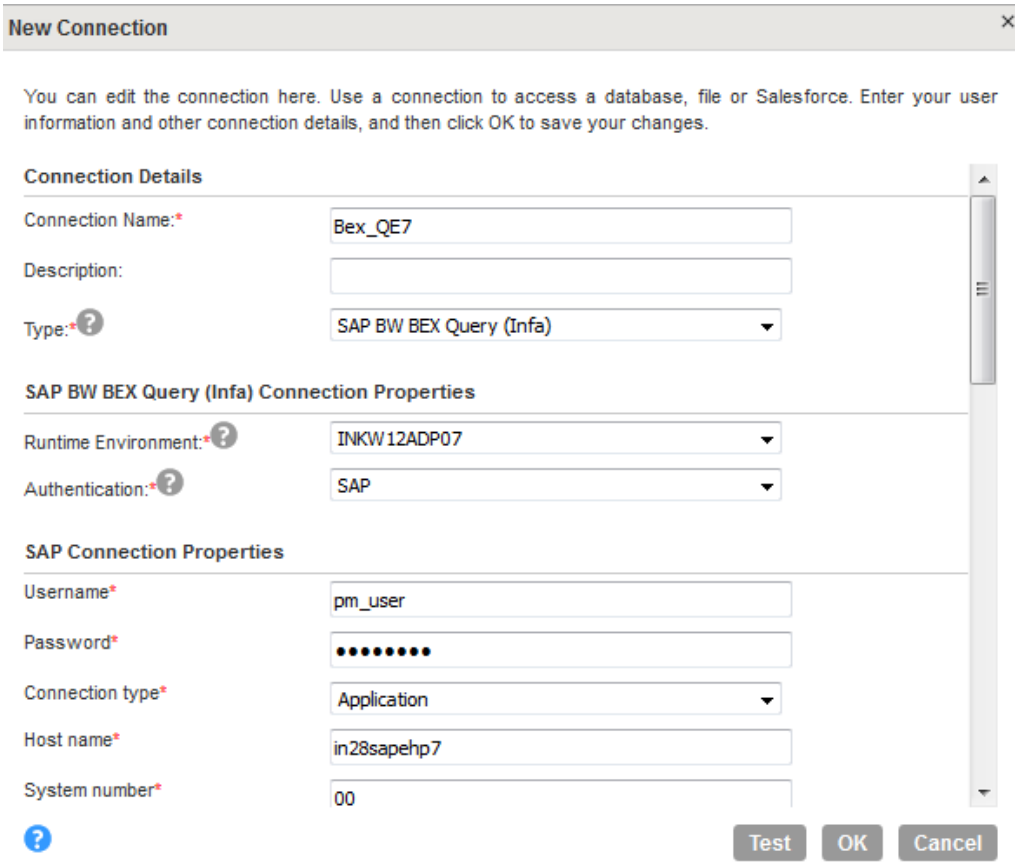
You can view the HTTP and HTTPS log files in the SMICM transaction. Optionally, you can increase trace level to 3 to view the detailed logs.

Mapping and mapping task example with SAP BW BEx Query

You are a financial consultant and you want to use SAP BW BEx Query Connector to read data from an SAP BW BEx query that extracts a financial planning report in SAP BW. You want to configure a filter to extract a specific project plan from the financial report.

To configure a mapping and mapping task to extract a specific project plan from SAP BW BEx Query and write to a flat file, perform the following tasks:

1. Create an SAP BW BEx Query connection. Select `SAP` as the authentication method, select `Application` as the connection type, and enter the SAP connection properties.
   The following image shows the configured SAP BW BEx Query connection:

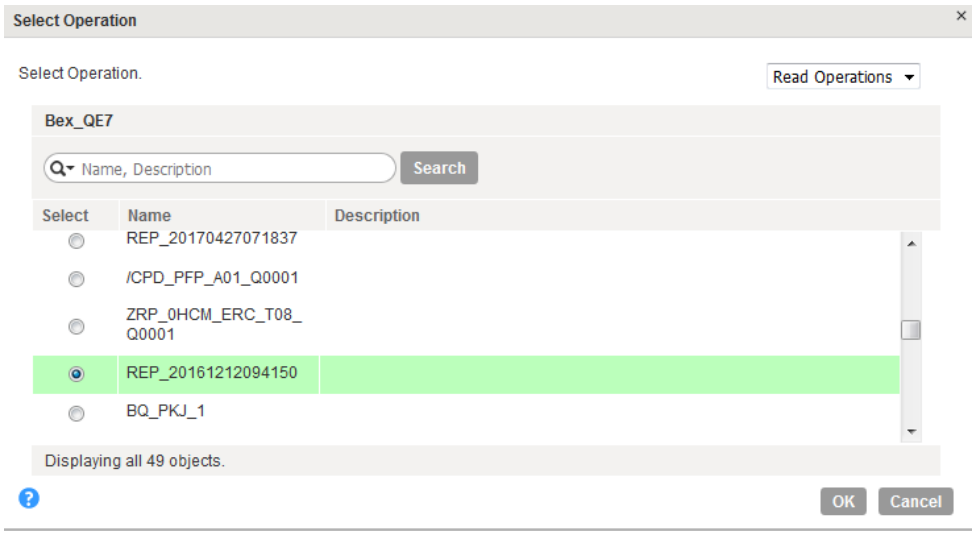


2. Create a flat file connection to write data to the flat file.
3. Create an SAP BW BEx Query mapping in the Cloud Mapping Designer.
   The following image shows the SAP BW BEx Query mapping:



4. Add a Source transformation, specify a name and description in the general properties.
5. On the **Source** tab, perform the following steps:
   a. In the **Connection** field, select the configured SAP BW BEx Query connection to connect to SAP BW BEx Query.
   b. In the **Operation** field, select query `REP_20161212094150` as the operation to extract the financial planning report from SAP BW BEx Query.

The following image shows the queries from the list of read operations:



You can view the SAP BW BEx source and flat file target in the mapping.

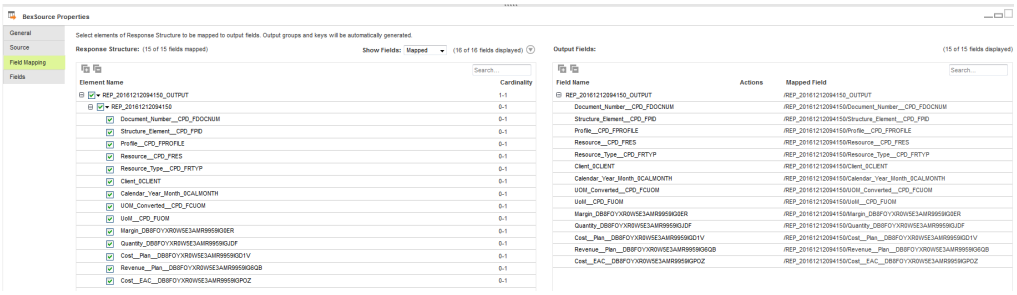
c. In the **Request Options** section, configure the request message in the following XML format, specify the attributes in the message, and validate the message:

```
<!--1 or more repetitions:-->
<proc:REP_20161212094150_INPUT xmlns:proc="http://xml.schemas/infa/procedure/">
  <!--Optional:-->
  <REP_20161212094150>
    <!--Optional:-->
    <var_temp>$$Param_variable</var_temp>
  </REP_20161212094150>
</proc:REP_20161212094150_INPUT>
```

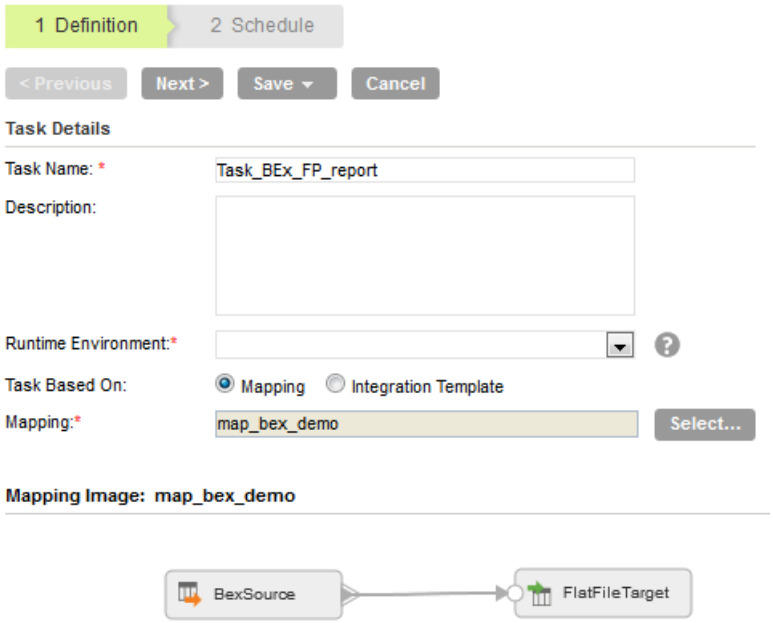
The request message specifies the criteria to include the financial planning report query `REP_20161212094150` and includes a parameter to filter a particular project plan within the financial planning report. You need to define the parameter `$$Param_variable=PR00004446` entry in a parameter file, `parameterfile_name.txt`, and save the file in the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/data/userparameters`. Later, enter the parameter file name on the Runtime Options and Advanced Options page of the Mapping Task wizard.

d. In the **Advanced Properties** section, set the tracing level to Normal, and use the default cache size of 1024.

6. On the **Field Mapping** tab, map all descendants from the incoming fields of the SAP BEx query response structure to the output fields.
The following image shows the response structure on the left pane and the output fields on the right pane:



7. Add a flat file Target transformation and specify the target object for the transformation. Perform the following steps:
   a. Select a flat file connection for the target transformations to write data to the flat file.
   b. Draw a link to connect the previous transformation to the Target transformation.
   c. Create a target object `Bex_query.csv` file to write data to the target.
   d. Map the fields that you want to link to the target fields.
8. Click **Save and New Mapping Task** to save and create a mapping task.
   The following image shows the Mapping Task wizard with the applied mapping:



9. Under **Advanced Options** on the **Runtime Options** tab, specify the parameter file name `parameterfile_name.txt` in which you configured the filter parameter value for the request XML
10. When you save and run the mapping, the Secure Agent retrieves the specified project plan from SAP BW BEx Query and writes the data to the corresponding flat file.
    The target flat file shows the following data extracted from the specified project plan after you run the task:

```
"Document_Number__CPD_FDOCNUM","Structure_Element__CPD_FPID","Profile__CPD_FPROFILE",
"Resource__CPD_FRES","Resource_Type__CPD_FRTYP","Client_0CLIENT",
"Calendar_Year_Month_0CALMONTH","UOM_Converted__CPD_FCUOM","UoM__CPD_FUOM","Margin_DB
8FOYXR0W5E3AMR9959IG0ER","Quantity_DB8FOYXR0W5E3AMR9959IGJDF",
"Cost__Plan__DB8FOYXR0W5E3AMR9959IGD1V","Revenue__Plan__DB8FOYXR0W5E3AMR9959IG6QB","C
ost__EAC__DB8FOYXR0W5E3AMR9959IGPOZ","Var_Temp"
,"800PR00004446",,"800ZMRS0MRS1429","800ZMRS0MRS",,,,,-12,0,1200,0,20000,"PR00004446"
```

Note: You can also read data from an SAP BW BEx query by using a Web Services transformation. For more information, see the Informatica How-To Library article, "Configuring a Web Services Transformation in Informatica Cloud Data Integration to Read Data from an SAP BW BEx Query". To get the Informatica How-To Library article, contact Informatica Global Customer Support.

Rules and guidelines for SAP BW BEx Query operations

Consider the following rules and guidelines when you read data from SAP BW BEx Query:

- To read large number of records from SAP BW BEx Query, you must perform the following tasks:
  - Set the JVM options for type DTM to increase the -Xms and -Xmx values in the system configuration details of the Secure Agent.
  - Increase the cache size for the web service response in the advanced properties of the SAP BW BEx Source transformation
- You can configure characteristics to be displayed as **Key and Text**, **Key**, or **Text and Key**. To do this, open the query in the SAP BEx Query Designer, select the characteristic, and click the **Display** tab on the right pane. In the **Value Display** section, select the **Key and Text**, **Key**, or **Text and Key** option under **Display As**. In the **Result Rows** section, select the **Always Suppress** option.
- When you configure a mapping to read data that contains the Date and Time data types for key figures from an SAP BW BEx query, you must map the Date and Time data types of key figures to String.

Limitations for SAP BW BEx Query operations

SAP BW BEx Query Connector has the following known limitations:

- You cannot extract a large volume of data in SAP BW BEx Query mappings.
- You cannot read a large number of columns and rows. You can read less than 10 to 15 columns and 45,000 rows without performance degradation.
  To know more about the performance details of SAP BW BEx Query Connector, see KB article 000160936.
- You cannot read data from the SAP BW BEx query that contains more than one key figure structure. The value of each key figure must be in a single currency.
- You cannot read data from free characteristics. Even though the free characteristic fields appear in the response structure, the data from the free characteristics is not fetched in the output.
- You cannot read data from the SAP BW BEx query that contains the following components:
  - Characteristic hierarchies
  - Characteristic structures
- You cannot perform an SAP BW BEx query data extraction when the technical name of the variable in the BEx Query Designer exceeds eight characters.
- You can only configure characteristics as rows and key figures as columns in the SAP BEx Query mapping.

APPENDIX A

SAP data type reference

Data Integration uses the following data types in mappings and mapping tasks with SAP:

Native data types

Native data types are data types specific to the source and target databases or flat files. They appear in non-SAP sources and targets in the mapping.

SAP data types

SAP data types appear in the Fields tab for Source and Target transformations when you choose to edit metadata for the fields. SAP performs any necessary conversion between the SAP data types and the native data types of the underlying source database tables.

Transformation data types

Set of data types that appear in the remaining transformations. They are internal data types based on ANSI SQL-92 generic data types, which Data Integration uses to move data across platforms. Transformation data types appear in all remaining transformations in a mapping or mapping task.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

SAP BW BEx Transformation Data Types

The following table lists the SAP data types that SAP BW BEx Query Connector supports along with the corresponding transformation data types:

SAP Data Type	Transformation Data Type	Range for Transformation Data Type
CHAR	String	1 to 104,857,600 characters Fixed-length or varying-length string.
CUKY	String	1 to 104,857,600 characters Fixed-length or varying-length string.
CURR	Decimal	Precision 1 to 28 digits, scale 0 to 28
DATS	Date/time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision to the nanosecond.
DEC	Double	Precision 1 to 28 digits, scale 0 to 28

SAP Data Type	Transformation Data Type	Range for Transformation Data Type
FLTP	Double	Precision 15, scale 0
INT4	Integer	Precision 10, scale 0
LANG	String	1 to 104,857,600 characters Fixed-length or varying-length string.
NUMC	String	1 to 104,857,600 characters Fixed-length or varying-length string.
QUAN	Decimal	Precision 1 to 28 digits, scale 0 to 28
TIMS	Date/time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision to the nanosecond.
UNIT	String	1 to 104,857,600 characters Fixed-length or varying-length string.

INDEX

C

Cloud Application Integration community
  URL 4
Cloud Developer community
  URL 4

D

Data Integration community
  URL 4
data types
  SAP 24

I

Informatica Global Customer Support
  contact information 5
Informatica Intelligent Data Management Cloud
  web site 4

M

maintenance outages 5
mapping example
  Source transformation 19

S

SAP
  data types 24
SAP BW BEx mapping tasks
  configuring 16
SAP BW BEx mapping tasks *(continued)*
  creating 18
  monitoring 19
  overview 15
SAP BW BEx mappings
  configuring 16
  configuring the Source transformation 16
  overview 15
  selecting a read operation 16
SAP BW BEx Query
  assets 6
SAP BW BEx query connection
  properties 9
SAP integration methods
  using SAP BW BEx Query 6
status
  IDMC 5
system status 5

T

trust site
  description 5

U

upgrade notifications 5

W

web site 4